Course: Management of AI products

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LECTURE REFLECTIONS for sessions 1 to 7

- **Lecture 1**
  - Product manager owns the product problem and the outcomes of all the decisions involved for users and business
  - Product managers in general are accountable to make trade-offs and judgement calls to get the desired outcomes
  - Alignment, authority and escalation are to be used by PM as most of the time, authority is absent; PM is the CEO of the product
  - Everyone involved in the project needs to contribute but then the PM owns the recommendation and unresolved risk-escalation
  - PM has a lot of responsibilities: understanding users, competitors, markets, business model; defining product vision, strategy, outcome; prioritizing opportunities and sequence learning; aligning delivery, launch, adoption, measurement, operation and product evolution
  - PM does this regularly: customer and market learning, decision work, teamwork, operating work
  - Skill stack needed by PM: customer and product judgement, business judgement, technical and data fluency, leadership
  - Product management is challenging due to: high accountability with incomplete information and constrained formal authority, conflicting demands, delayed feedback and emotional load

- **Lecture 2**
  - Product management is enriching too: turn problems into products with value, get cross-learning opportunities, work with talented specialists, go from hypothesis to adoption to evolution
  - Product lifecycle is a cycle: discover-define-strategize-plan-build-launch-grow-retire
  - Discover – identify a problem worth solving; define and strategize – choose where to play; plan and prioritize – sequence value and learning; build and learn – solve with product team; launch and availability – product/operations/market ready; grow and operate – compound value; retire, replace, renew
  - PM artifact system – this is needed to preserve decisions, evidence, ownership and change history (artifact types – discovery, strategy, business, delivery, launch, measurement, risk, learning
  - Tracing one decision across artifacts is needed and frameworks are needed for decision making
  - Frameworks:

- JTBD - Customers do not buy products because of features. The real job is not the product. The real job is the progress the customer seeks. JTBD is a customer-centric framework that identifies the progress customers seek in a specific situation. It focuses on the job customers want accomplished rather than product features or demographic characteristics.
- Double Diamond - describes how successful teams move from problem space (discover + define) to solution space (develop + deliver); key concept is to - first diamond = find the right problem & second diamond = find the right solution; each diamond has divergence and convergence phases.
- Opportunity Solution Tree is a visual decision-making framework that helps teams move from: business outcome to customer opportunities to possible solutions to assumption tests to evidence; Instead of jumping directly to features, OST forces you to understand: What outcome are we trying to achieve? What customer opportunities exist? Which solution best addresses those opportunities? What evidence do we have?
- 3 frameworks together: Lean Canvas → Is this idea worth pursuing? Business Model Canvas (BMC) → How will this business create and capture value? Product-Market Fit (PMF) Pyramid → Does the product solve a real customer need? Lean Canvas focuses on the product idea. BMC focuses on the entire business ecosystem. PMF Pyramid tests alignment from target customer and underserved needs through value proposition, feature set and user experience.
- Product Frameworks: What product should we build? Strategy Frameworks: Where should we compete and how should we win?
- PESTLE – Political, Economic, Social, Technological, Legal, Environmental: PESTLE helps analyze macro-environmental factors outside the organization's control.
- Porter's 5-forces – Competitive rivalry, threat of new entrants, bargaining power of buyers, bargaining power of suppliers, threat of substitutes.
- SWOT – Strengths (Internal), Weakness (Internal), Opportunities (External), Threats (External); SWOT - Collects information. TOWS - Creates actions. SO Strategy - Use strengths to exploit opportunities. WO Strategy - Use opportunities to overcome weaknesses. ST Strategy - Use strengths to counter threats. WT Strategy - Reduce weaknesses and avoid threats.
- Ansoff Matrix - helps determine how to grow. Market penetration, product development, market development and diversification.
- BCG Matrix - helps decide where to invest resources. Star - High growth High market share; Cash Cow - Low growth High market share; Question Mark - High growth Low market share; Dog - Low growth Low market share.
- Three Horizons Framework: Horizon 1 – core business, horizon 2 – emerging growth, horizon 3 – future bets
- Lifecycle fit: PESTLE → What's changing? Five Forces → How tough is the market? SWOT → What is our position? TOWS → What should we do? Ansoff → How should we grow? BCG → Where should we invest? Horizons → How do we balance today and tomorrow?

- Quality Function Deployment (QFD) is a structured method for converting: Customer Requirements (WHATs) into Technical Requirements (HOWs); The primary tool used in QFD is called the House of Quality because its shape resembles a house.
- 3 complementary frameworks: North Star Metric (NSM) → Are we creating value? (single metric that best represents the value delivered to customers); AARRR Funnel → Where are users dropping off? (locates a lifecycle conversion constraint – acquisition, activation, retention, revenue, referral); HEART Framework → How is the user experience? (translates experience goals into signals and metrics – happiness, engagement, adoption, retention, task success); GSM – Goal, signal, metric
- Kano: Understand customer satisfaction impact; 5 categories: must-be features, performance features, delighter, indifferent, reverse; RICE: Prioritize among options; It helps compare competing feature ideas objectively. MoSCoW: Define release scope; must-have, should-have, could-have, won't-have
- What transfers across product contexts: user outcome, baseline, evidence, owner, revision trigger
- What changes across markets and products: workflow and consequences, language and data, decision rights, economics and operations
- Adapt frameworks without cargo culting: Cargo culting means: Copying a framework, process, or best practice blindly without understanding why it worked elsewhere. Source Mechanism => Context Audit => Local Evidence => Adapted Decision => Reversal Trigger
- When decisions must be made under uncertainty, who decides, how fast should we decide, and how do we avoid paralysis:

  Frameworks:

  - RACI – responsible, accountable, consulted, informed (who performs work)
  - DACI – driver, approver, contributors, informed (who makes decision)
  - RAPID – recommend, agree, perform, input, decide (complex cross-functional decision)
- One-Way Door Decisions - Hard to reverse, expensive to undo
- Two-Way Door Decisions – Easy to reverse
- Cynefin helps teams act when cause and effect are uncertain.
- OODA – Observe, orient, decide, act
- RACI: Defines responsibilities. DACI: Identifies approver. RAPID: Handles technology selection. One-Way Door: Choosing vendor architecture. Two-Way Door: Prompt optimization. Cynefin: Treat user adoption as a complex problem. Run experiments. OODA: Continuously improve based on user feedback.

- What AI PM owns:-
  - Directly owns - problem framing, success logic, prioritization, launch criteria
  - Co-owns - workflow quality, risk trade-offs, escalation paths
  - Influences - model choices, data choices, platform choices
- The AI PM must expose ownership gaps before launch pressure hides them.

        - For each major trade-off, identify: Decider -> Reviewer -> Escalation owner
    - A central AI-PM skill is translation. Business stakeholders need to understand:
        - latency + quality + data limitations as product realities.
    - Think of the stack as:
        - Data -> Model -> API/Application Logic -> UX
        - **Layer** **Evidence**
        - Data freshness
        - Model model lift
        - API reliability
        - UX adoption
    - AI PM work requires alignment across:
        - Product + Design + Engineering + Data Science + Legal/Business
        - The problem is that different functions can use the same words - "quality", "safety", "readiness" – while meaning different things. Therefore the PM has to make trade-offs explicit and document them.
        - Evidence - handoff quality, rework rate, late-stage conflict, documented trade-offs.
    - When quality, speed, and cost disagree, which trade-off should the PM defend and why?
        - Evidence should include latency by workflow, cost per successful task, accuracy by segment, business impact of degraded performance.

- **Lecture 3**
    - Real AI opportunities come from workflow leverage, not feature novelty.
    - The strong opportunities are those that: improve recurring decisions, reduce friction, unlock scale.
    - PM should examine:
        - pain + frequency + measurability + defensibility, rather than AI hype.
    - A strong opportunity needs both: Value and Feasibility
    - The two classic mistakes are:
        - Easy but low value - The organization picks something technically convenient that does not matter enough.
        - Valuable but unrealistic - The opportunity is attractive but has no plausible execution path.
    - Build, Buy, Partner, API –
      The evidence includes:
        - strategic importance
        - switching cost
        - vendor dependency
        - Compliance fit
        - time-to-market
        - ownership of core IP.

- The classic failures are to build what should be bought or buy what should be differentiated or ignore vendor/governance/margin implications.
- A platform claim must be earned. Evidence for platform expansion includes:
  - cross-workflow applicability
  - ecosystem pulls
  - extensibility
  - repeat use
  - clear adoption in the first use case
- JTBD is one of the key conceptual anchors. The central question: What progress is the customer trying to make, and where does AI reduce effort, anxiety, or delay?

- **Lecture 4**
  - AI-specific Product Market Fit lens is Value + Trust + Repeat Use; A product can have enormous curiosity/trial but still lack durable PMF.
  - ChatGPT as a platform: Evidence includes reuse across jobs, ecosystem participation, partner leverage, user habit formation, extensibility beyond one task.
  - Recommendation Systems: Is the recommendation improving discovery, increasing basket/business value, increasing retention, or simply reducing catalog overload?
    - Relevant evidence: engagement depth, conversion lift, basket size, retention impact, whether recommendations feel trusted instead of intrusive.
  - What would make this AI product hard to replace once users understand its value?
    - Data advantage + Workflow fit + Switching cost + Brand trust + Distribution
  - Opportunity mapping - The PM should examine workflow + why-now + capability choice + funding/rejection decision and require evidence such as:
    - market pull
    - data readiness
    - distribution logic
    - reason this move should happen now rather than another option.

- **Lecture 5**
  - UX for AI products - The fundamental decision is: Should AI be =>
    - visible
    - invisible
    - optional
    - interruptive

        - assistant-like
        - inline
        - background automation
    - Failure modes
        - Too visible: AI fatigue
        - Too invisible: user confusion and distrust
        - Evidence applicable here is:-
            - adoption
            - task completion
            - override behavior
            - trust signals
            - whether users know when AI is helping
    - Design for uncertainty (for AI products)
        - Speak confidently
        - Hedge
        - Ask for clarification
        - Return control to user
        - Evidence applicable here is:-
            - correction rate
            - abandonment
            - clarification behavior
            - whether users handle uncertainty safely
    - HITL – Human in the loop
        - Where does human review add judgment, trust, or harm reduction and where does it merely add: cost + delay + operational burden?
        - The HITL framework includes:
            - routing thresholds
            - reviewer responsibilities
            - turnaround targets
            - escalation logic
            - override behavior
    - Prompt design includes:
        - Role + Context + Task + Constraints and should be documented with:
            - sample inputs
            - blocked instructions
            - retrieval boundaries
            - Fallback behavior
    - Explainability, trust and transparency:
        - The material ties explainability to:
            - user-facing explanation
            - source citation/grounding

- trust controls
- safe recovery
- avoiding deceptive anthropomorphism

- **Lecture 6**
  - Data quality is product quality. A broken data foundation can make a strong model useless.
    - Evidence:
      - event completeness
      - freshness
      - cohort coverage
      - downstream feature availability
  - We should log data that improves:
    - product decisions
    - ranking
    - safety outcomes
  - Data collection failure has two bad extremes:
    - Over-collection - Creates privacy burden
    - Under-collection - Creates blind spots no model can recover from
  - The label is not merely clerical metadata. It is a product decision because the label determines what the system is learning. Types of labels - expert labels, proxy labels, crowdsourced labels.
  - Data Pipeline : Ingestion => Storage => Transformation => Feature pipeline => Serving => Feedback Loop
    - Critical evidence:
      - pipeline latency
      - failed jobs
      - feature freshness
      - training-serving skew
  - A core AI PM skill is translating business problems into ML problems:
    - outcome definition
    - input features
    - prediction target
    - evaluation setup

- **Lecture 7**
  - AI Lifecycles differ from software-only lifecycles because deployment is not completion.
  - The ownership challenge increases after release because:
    - data changes
    - behavior changes
    - model performance changes
    - workflows change
    - business conditions change.
  - The lifecycle requires: launch gate + post-launch checks + drift alerts + response ownership
  - The key lifecycle gates are ideation, discovery, validation, launch, monitoring, and iteration.
  - Lifecycle deliverables artifact stack: experiment brief => launch checklist => model card => drift review.
  - A mature AI product needs to remain reliable under change, so AI PM needs to plan for:
    - Monitoring
    - Drift
    - Incident reviews
    - Escalation rules
    - Response ownership